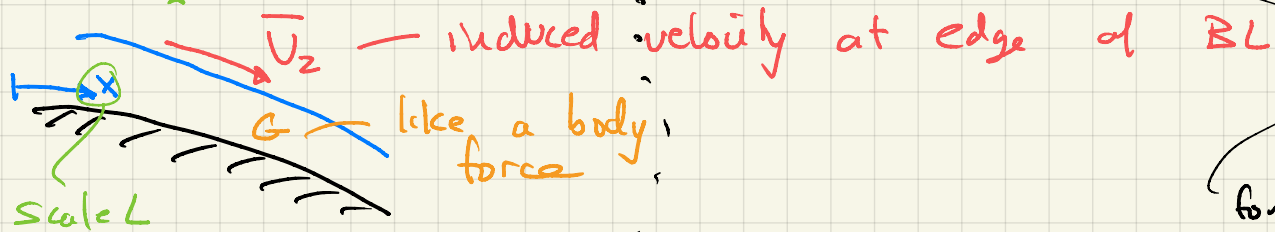


Lecture 28

- Steaming III (and last)

last time...

$$U_i = U_{\text{entr}} \quad \bar{U}_2 = \frac{3}{8n} \left[- \frac{d\hat{U}\hat{U}^*}{dx} + \boxed{i} \left(\hat{U}^* \frac{d\hat{U}}{dx} - \hat{U} \frac{d\hat{U}^*}{dx} \right) \right]$$



form $z - z^*$

$\in \text{Imaginaires}$

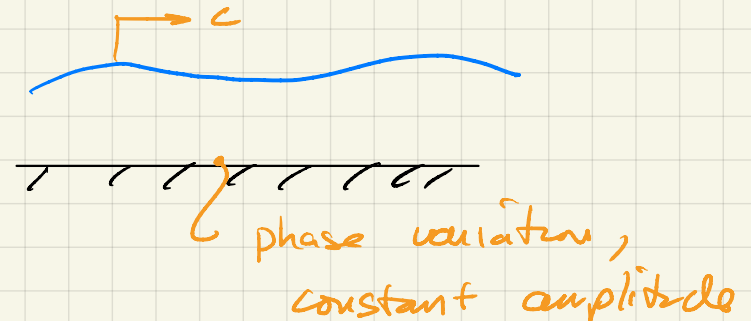
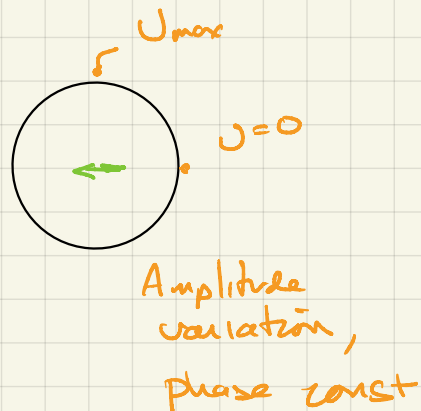
$\boxed{x\dot{U}} \rightarrow \in \text{Reals}$

$$\hat{U}\hat{U}^* = |\hat{U}|^2$$

$\in \text{Reals}$

- $\hat{U}(x)$ is the main driver

→ understand role of Amplitude and Phase



$$\bar{U}_2 = \frac{3}{8n} \left(-\frac{d\hat{U}\hat{U}^*}{dx} + i \left(\hat{U}^* \frac{d\hat{U}}{dx} - \hat{U} \frac{d\hat{U}^*}{dx} \right) \right)$$

$$\hat{U}(x) = A(x) e^{i\phi(x)}$$

↑
real
↑
real

$$\hat{U}\hat{U}^* = A^2$$

$$\hat{U}^* \frac{d\hat{U}}{dx} - \hat{U} \frac{d\hat{U}^*}{dx} = A e^{-i\phi} \frac{d}{dx} (A e^{i\phi}) - A e^{i\phi} \frac{d}{dx} (A e^{-i\phi})$$

$$= A \left[\cancel{e^{-i\phi} \frac{dA}{dx} e^{i\phi}} + \cancel{e^{-i\phi}} i A \frac{d\phi}{dx} \cancel{e^{i\phi}} - \cancel{e^{i\phi} \frac{dA}{dx} e^{-i\phi}} + \cancel{e^{i\phi}} i \frac{d\phi}{dx} A \cancel{e^{-i\phi}} \right]$$

$$= 2A^2 i \frac{d\phi}{dx}$$

$$\bar{U}_2 = -\frac{3}{8n} \left[\frac{dA^2}{dx} + 2A^2 \frac{d\phi}{dx} \right]$$

↑
due to
amplitude
↑
due to
phase

NOTE: Why no viscosity in $\bar{U}_z$ expression?

→ still in BL limit → a $Re \rightarrow \infty$ model
for large Re

• Do the particles move at the local flow velocity $\bar{U}_z$?

↑ like in visualization —
a piece of mass in flow

— No ... subtle difference

— e.g.

$\underline{a}$

↑
acceleration
of a
fluid particle

$\neq$

$\frac{\partial \underline{u}}{\partial t}$

↑
change in velocity
at a point

field "Eulerian"
fixed in space
 $\underline{u}(\underline{x}, t)$

$$\underline{a} = \frac{D\underline{u}}{Dt} = \frac{\partial \underline{u}}{\partial t} + \underline{u} \cdot \nabla \underline{u}$$

—

$\underline{v}$

of a particle
of fluid

$\neq$

$\frac{\partial \underline{x}_p}{\partial t}$

$$\underline{U} = \frac{D \underline{x}_p}{Dt}$$

continuous track
of particle
locations

- Drift — streaming causes drift $\underline{w} \neq \underline{u}$

J. Fluid Mech, Vol 1 "Drift"
Lighthill

define $\underline{w}(\underline{x}_0, t)$ — velocity of fluid particle at $\underline{x}_0$
at time t_0

$\underline{u}(\underline{x}, t)$ — "normal" velocity field

$$\underline{w}(\underline{x}_0, t) = \underline{u} \left(\underline{x}_0 + \int_{t_0}^t \underline{w}(\underline{x}_0, \tau) d\tau, t \right)$$

accounts for the
 $\underline{x}_0$ "labeled" particle moving...

short times $t - t_0$ small so $|\underline{x} - \underline{x}_0|$ small

so $\underbrace{\int_{t_0}^t \underline{w}(\underline{x}_0, t) dt}_{\text{like } \varepsilon}$ small
 ... Taylor series it

$$\underline{w}(\underline{x}_0, t) = \underline{u}(\underline{x}_0, t) + \underbrace{\left(\int_{t_0}^t \underline{w}(\underline{x}_0, t) dt \right)}_{\varepsilon} \cdot \nabla \underline{u} \Big|_{\underline{x}_0} + O(t - t_0)^2$$

$$\underline{w} \approx \underline{u} + O(t - t_0)$$

$$\int_{t_0}^t O(t - t_0) dt = O(t - t_0)^2$$

$$\underline{w}(\underline{x}_0, t) = \underline{u}(\underline{x}_0, t) + \left(\int_{t_0}^t \underline{u}(\underline{x}_0, t) dt \right) \cdot \nabla \underline{u} \Big|_{\underline{x}_0} + O(t - t_0)^2$$

apply at $y = \delta$

$$\overline{w}_z(\underline{x}_0) = \overline{U}_z(\underline{x}_0) + \underbrace{\operatorname{Re} \left\{ \underbrace{\left(\int_{t_0}^t U_1(\underline{x}_0) dt \right)}_{\varepsilon} \underbrace{\frac{dU_1}{d\underline{x}}}_{\varepsilon} \right\}}_{\varepsilon^2}$$

$$U_1 = \frac{1}{2} \{ \hat{U} e^{i\omega t} + \hat{U}^* e^{-i\omega t} \}$$

$$\int_{t_0}^t U_1 dt = \frac{1}{2i\omega} \{ \hat{U} e^{i\omega t} - \hat{U}^* e^{-i\omega t} \}$$

$$\frac{dU_1}{dx} = \frac{1}{2} \left\{ \frac{d\hat{U}}{dx} e^{i\omega t} + \frac{d\hat{U}^*}{dx} e^{-i\omega t} \right\}$$

all $e^{i\omega t}$ terms have for $e^{i\omega t} e^{-i\omega t}$

which time average to zero

$$\overline{e^{i\omega t} e^{-i\omega t}} = 0$$

$$\bar{W}_2 = \bar{U}_2 + \underbrace{\frac{1}{4i\omega} \left\{ \hat{U} \frac{d\hat{U}^*}{dx} - \hat{U}^* \frac{d\hat{U}}{dx} \right\}}_{\text{as above for } \bar{U}_c} + \underbrace{\left\{ e^{\pm 2i\omega t} \text{ terms} \right\}}_{\rightarrow 0}$$

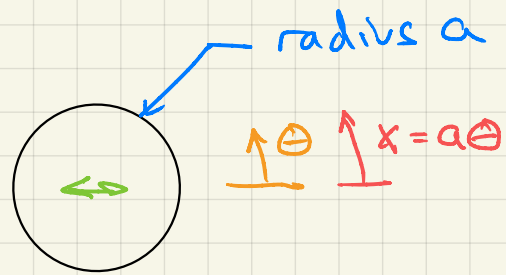
$-\frac{3}{8n} \frac{dA^2}{dx} - \frac{3}{4n} A^2 \frac{d\phi}{dx}$

$-\frac{A^2}{2n} \frac{d\phi}{dx}$

$$= -\frac{3}{8n} \frac{dA^2}{dx} - \frac{5}{4n} A^2 \frac{d\phi}{dx}$$

only ϕ phase affects drift

Revisit



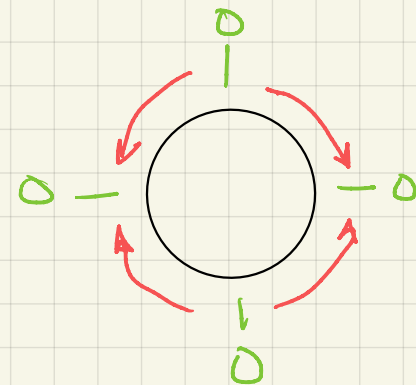
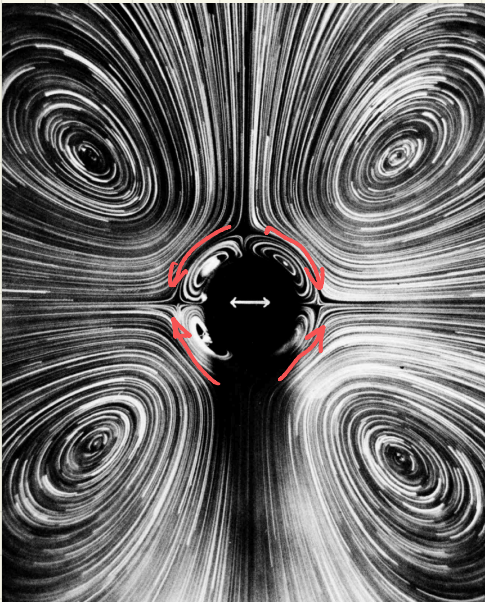
$\hat{U} = 2U_0 \sin \theta = A$
potential flow
solution on
surface
(edge of BL)

$$\omega_z = -\frac{3}{8n} \frac{dA^2}{dx} \quad x = a\theta$$

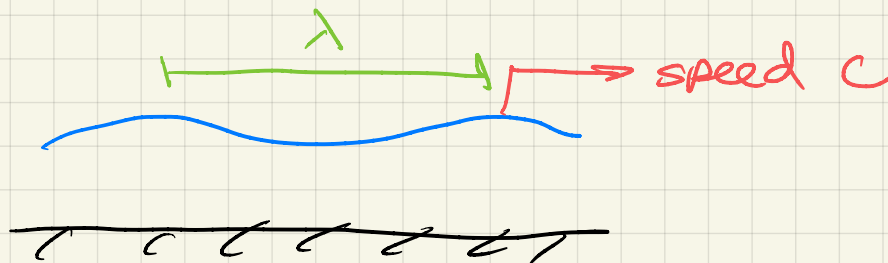
$$\omega_z = -\frac{3}{8n} \frac{1}{a} \frac{d}{d\theta} [4U_0^2 \sin^2 \theta]$$

$$= -\frac{3}{2n} \frac{1}{a} U_0^2 \underbrace{2 \sin \theta \cos \theta}_{\sin 2\theta}$$

$$= -\frac{3}{2} \frac{U_0^2}{an} \sin 2\theta$$



Progressive wave



$$\hat{U} = A e^{i\phi}$$

A const

$$\phi = -\frac{2\pi x}{\lambda}$$

$$\bar{U}_2 = -\frac{2}{\rho_m} A^2 \frac{d\phi}{dx}$$

$$= \frac{3}{4\rho_m} A^2 \frac{2\pi}{\lambda}$$

$\frac{1}{c}$

frequency

ang. freq

$$f = \frac{\omega}{2\pi}$$

$$c = \lambda f = \frac{\lambda \omega}{2\pi}$$

$$\bar{U}_2 = \frac{3}{4\rho_m} A^2 \frac{1}{c}$$

$$\bar{W}_2 = \frac{5}{4\rho_m} A^2 \frac{1}{c}$$

A small

$$\bar{W}_2 \ll c$$